

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0037

Norrie 138 kV
July 23, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0037. The Interconnection Customer proposes a 171.8 MW solar photovoltaic facility interconnecting at the Norrie 138 kV substation in the Virginia area. The request was received on May 19, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 32 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$53,615,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Brush Tavern-Yaphank 138kV: 113.8% loading (311 MVA) following the outage of the Cristianitos-Enslen Substation 138kV; DFAX = 8.0%.

For the loss of Bear-Clover 345kV L, the Bear-Mecklenburg Power Station 345kV loads to 107.9% of its 1,927 MVA rating. Project DFAX: 23.0%.

Bear-Leavitt 345kV: 107.2% loading (1,580 MVA) following the outage of the Bear-Leavitt 345kV Cir. 2; DFAX = 16.9%.

For the loss of Reusens Substation 138 kV/1 kV Transformer, the Flowood-Unity 138kV L loads to 106.9% of its 252 MVA rating. Project DFAX: 8.3%.

Mecklenburg Power Sta-Person 345kV L: 106.7% loading (2,318 MVA) following the outage of the Clover-Bear 345kV; DFAX = 32.4%.

Clover-Bear Substation 345kV: 106.3% loading (2,416 MVA) following the outage of the Person-Mecklenburg Power Sta Substation 345kV line; DFAX = 34.3%.

Fluvanna P.S-Louisa Generation Facility 345kV: 106.0% loading (1,711 MVA) following the outage of the Leaitt-Birchwood Substation 345kV Line; DFAX = 18.2%.

Littlefork-Lake Lure 138kV: 104.8% loading (494 MVA) following the outage of the Louisa Generation Facility-Fluvanna P.S 345kV L; DFAX = 6.7%.

The Bluff City-Lake Lure 138kV is overloaded to 103.4% of its 408 MVA emergency rating for the loss of the Fluvanna P.S-Louisa Generation Facility 345kV Line. The project contributes 6.5% DFAX (11.2 MW) to this overload.

Person-State School 345kV: 102.8% loading (1,605 MVA) following the outage of the Mecklenburg Power Sta-Person 345kV; DFAX = 20.5%.

The South Coltn-Flowood 138kV is overloaded to 102.7% of its 178 MVA emergency rating for the loss of the Belagio-Otter River 138kV. The project contributes 6.9% DFAX (11.8 MW) to this overload.

Unity-Wythe 138kV: 102.4% loading (244 MVA) following the outage of the Reusens 138-1 kV TX; DFAX = 8.3%.

The Franklin-Elk Grove-S Texa Substation 138kV is overloaded to 102.0% of its 168 MVA emergency rating for the loss of the Desert Outpost-South Texa 138kV. The project contributes 6.5% DFAX (11.1 MW) to this overload.

The Louisa Generation Facility-Carrabelle 345kV is overloaded to 100.2% of its 1,492 MVA emergency rating for the loss of the N Anna-Louisa Generation Facility 345kV. The project contributes 10.8% DFAX (18.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yaphank-Brush Tavern 138kV	Rebuild line with higher-capacity conductor	\$4,187,000	15% (\$636,000)
Bear-Mecklenburg Power Station 345kV	Reconductor line	\$51,413,000	18% (\$9,148,000)
Leavitt-Bear 345kV Cir. 1	Reconductor line	\$70,102,000	18% (\$12,241,000)
Flowood-Unity 138kV	Reconductor line	\$3,591,000	15% (\$533,000)
Mecklenburg Power Sta-Person 345kV L	Reconductor line	\$31,461,000	18% (\$5,604,000)
Bear-Clover 345kV Line	Reconductor line	\$20,042,000	18% (\$3,564,000)
Fluvanna P.S-Louisa Generation Facility Substation 345kV L	Reconductor line	\$32,383,000	17% (\$5,574,000)
Littlefork-Lake Lure 138kV	Reconductor line	\$1,934,000	17% (\$326,000)
Bluff City-Lake Lure 138kV	Reconductor line	\$5,533,000	17% (\$916,000)
State School-Person 345kV	Reconductor line	\$15,831,000	18% (\$2,851,000)
South Coltn-Flowood 138kV	Reconductor line	\$1,644,000	15% (\$245,000)
Unity-Wythe Substation 138kV Line	Reconductor line	\$3,358,000	15% (\$499,000)
Franklin-Elk Grove-S Texa Substation 138kV	Reconductor line	\$5,143,000	17% (\$860,000)
Louisa Generation Facility-Carrabelle 345kV	Reconductor line	\$62,183,000	17% (\$10,618,000)
	Total allocated to project		\$53,615,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.